

MA3523

Thm 1.2 $\forall a, b \neq 0 \in \mathbb{Z}, \exists r, s \in \mathbb{Z}$ s.t. $\gcd(a, b) = ar + bs$

Def 1.4 $(G, *)$ is a Group if (1) $*$ is associative (2) identity element exists (3) inverse exists. $a + b \equiv (a + b) \pmod{n}$, $a \cdot b \equiv (a \cdot b) \pmod{n}$. $a|b \Leftrightarrow b = ak$, $b \equiv 0 \pmod{a}$, $b > a$

Thm 1.6 $a \neq 0 \in \mathbb{Z}$ and $\gcd(a, n) = 1 \Leftrightarrow a^{-1}$ exists $\pmod{n}$

Def 1.8 a Group of Units of $\mathbb{Z}_n$ is $(U(n), \cdot)$, $U(n) = \{a \in \mathbb{Z}_n | \gcd(a, n) = 1\}$

Def 1.9 a Permutation of A is a bijection $A \rightarrow A$

Def 1.10 a Symmetric Group on A is S_A , if $A = \{1, 2, \dots, n\}$, denote by S_n . **Thm 1.7.** $|S_n| = n!$

Def 1.11 Subgroup let G be a group, if $S \subset G$ is also a group under same operation, then S is subgroup of G

Thm 1.12 a subset H of G is a subgroup if (1) e in G is in H (2) closure (3) if $h \in H$, then $h^{-1} \in H$

Thm 1.13 H is a subgroup of G if $H \neq \emptyset$ and $\forall g, h \in H$, $gh^{-1} \in H$.

Thm 1.14 $\langle g \rangle = \{g^k | k \in \mathbb{Z}\}$ where $g \in G$ is a subgroup of G , and it is the smallest subgroup containing g .

Def 1.12 if $G = \langle g \rangle$, then it is Cyclic and g is the Generator. the Order of G is the smallest n , s.t. $g^n = e$. denote $|g| = n$. note: if $c \in G$, $\langle c \rangle$ is the smallest subgroup containing c in G .

Thm 1.17 the subgroups of $\mathbb{Z}$ is $n\mathbb{Z}$, $n \in \mathbb{Z}$

Thm 1.19 group $G = \langle a \rangle$ and $|G| = n$. if $b = a^k$, then $|b| = n / \gcd(k, n)$

Cor 1.2 generators of $\mathbb{Z}_n$ is r where $1 \leq r \leq n$ and $\gcd(r, n) = 1$

Def 2.3 a cycle of length 2 is called Transposition. $(a_1 a_2 a_3 \dots a_k) = (a_1 a_k)(a_1 a_{k-1}) \dots (a_1 a_3)(a_1 a_2)$

Def 2.4 the Alternating group $A_n = \{\text{all even permutation in } S_n\}$ is a subgroup of S_n **Prop 2.3** $|A_n| = n!/2$

Def 2.5 H is a subgroup of G . Left Coset $gH = \{gh | h \in H\}$ Right is similar

Thm 2.6 the cosets of H in G form a partition of G .

Lemma 2.2 let H be a subgroup of group G , following equivalent (1) $g_1 H = g_2 H$ (2) $H g_1^{-1} = H g_2^{-1}$ (3) $g_1 H \subset g_2 H$ (4) $g_2 \in g_1 H$ (5) $g_1^{-1} g_2 \in H$ **Thm 2.7** # left and right cosets equal

Prop 2.4 $g \in G$, $\phi : H \rightarrow gH$ is bijective (i.e. $|H| = |gH|$)

Def 2.6 the Index of H in G is the number of cosets of H in G , denote by $[G : H]$

Thm 2.8 (Lagrange) if G finite, then $[G : H] = |G|/|H|$ ($\Rightarrow |H| \mid |G|$)

Cor 2.1 $\forall g \in G, |g| \mid |G|$ **Cor 2.2.** if $|G|$ prime, then G is cyclic. **Cor 2.3.** $[G : K] = [G : H][H : K]$ (if $K \subset H \subset G$ and K, H are subgroups of G)

Def 3.1 two groups $(G, \cdot)$ and $(H, \circ)$ are Isomorphic if $\exists$ bijective map $\phi : G \rightarrow H$ s.t. $\phi(a \cdot b) = \phi(a) \circ \phi(b) \forall a, b \in G$, write $G \cong H$ and ϕ is an Isomorphism

Thm 3.1. if $\phi : G \rightarrow H$ is an isomorphism, then following true (1) $\phi^{-1} : H \rightarrow G$ is also an isomorphism (2) $|G| = |H|$ (3) both Abelian/cyclic or not (4) if G has a subgroup with order n , so is H

Thm 3.2 (Characterization of cyclic groups) All finite cyclic group $\cong \mathbb{Z}_n$, infinite cyclic groups $\cong \mathbb{Z}$

Thm 3.3 (Cayley's) every group $G \cong$ permutation group

Def 3.2 External Direct Product. two groups $(G, \cdot), (H, \circ), G \times H = \{(g, h) | g \in G, h \in H\}$ is a group with $(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 \circ h_2)$

Thm 3.4 let $(g, h) \in G \times H$. if $|g| = r, |h| = s$, then $|(g, h)| = \text{lcm}(r, s)$ **Thm 3.5.** $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ iff $\gcd(m, n) = 1$

Def 3.3 Let G be a group with subgroups H and K satisfying the following conditions. (i) $G = HK = \{hk : h \in H, k \in K\}$ (ii) $H \cap K = \{e\}$ (iii) $hk = kh$ for all $k \in K$ and $h \in H$ Then G is called the **internal direct product** of H and K .

Thm 3.6 If G is the internal direct product of subgroups H and K , then $G \cong H \times K$.

Def 4.1 a subgroup H of group G is Normal if $gH = Hg$ (denote $H \trianglelefteq G$)

Thm 4.1 following equivalent (1) $N \trianglelefteq G$ (2) $gNg^{-1} \subset N \forall g \in G$ (3) $gNg^{-1} = N \forall g \in G$

Thm 4.2 let $N \trianglelefteq G$, the cosets of N in G form a group under $(aN)(bN) = (ab)N \forall a, b \in G$

Def 4.2 if $N \trianglelefteq G$, the cosets of N in G is called factor group G and N (denote G/N , G quotient N) **Cor 4.4** $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$

Def 4.3. a group has no nontrivial normal subgroups is called Simple. **Thm 4.3** A_n is Simple if $n \geq 5$

Def 4.4 a Homomorphism of $(G, \cdot), (H, \circ)$ is a map $\phi : G \rightarrow H$ s.t. $\phi(g_1 \cdot g_2) = \phi(g_1) \circ \phi(g_2) \forall g_1, g_2 \in G$

Thm 4.4 let $\phi : G_1 \rightarrow G_2$ be a group homomorphism. then (1) $e_{G_2} = \phi(e_{G_1})$ (2) $\forall g \in G_1, \phi(g)^{-1} = \phi(g^{-1})$ (3) if H_1 is a subgroup of G_1 , so is $\phi(H_1)$ of G_2 (4) if H_2 is a subgroup of G_2 , so is $\phi^{-1}(H_2) = \{g \in G_1 | \phi(g) \in H_2\}$ (5) if $H_2 \trianglelefteq G_2$, then $\phi^{-1}(H_2) \trianglelefteq G_1$ (note: $\text{Ran} \phi$ is subgroup of G_2)

Def 4.5 let $\phi : G \rightarrow H$ be a group homomorphism. $\text{Ker} \phi = \{g \in G | \phi(g) = e_H\} \trianglelefteq G$. $\text{Ran} \phi = \{h \in H | \phi(g) = h, g \in G\}$

Lemma 4.5 a map $\phi : \mathbb{Z}_n \rightarrow \mathbb{Z}_m$ where $\phi(x) = ax$ for some $a \in \mathbb{Z}_m$ fixed. is a homomorphism iff $na \equiv 0 \pmod{m}$. numbers of homomorphism $\phi : \mathbb{Z}_n \rightarrow \mathbb{Z}_m$ is $d = \gcd(m, n)$ and are $a = (\frac{m}{d}k), k = 0, 1, \dots, d-1$

Thm 4.6 (Isomorphism I) if $\psi : G \rightarrow H$ is a homomorphism, then $\text{Ran} \psi \cong G/\text{Ker} \psi$

Thm 4.7 (Isomorphism II) if $N \trianglelefteq G$ and H is subgroup of G , then (1) HN is subgroup of G (2) $H \cap N \trianglelefteq H$ (3) $H/(H \cap N) \cong HN/N$

Thm 4.8 (Isomorphism III) let $N \trianglelefteq G, H \trianglelefteq G$ and $N \subset H$. then $G/H \cong (G/N)/(H/N)$ ($\varphi : G/N \rightarrow G/N$)

Def 5.1 A group G is called a p-group if every element of G has order is power of prime p . **Cor 5.1** $|G|$ is a p-group iff $|G| = p^n$.

Thm 5.1 (Fund. Thm. of Finite Abelian Groups). Every finite Abelian group G is isomorphic to $\mathbb{Z}_{p_1^{\alpha_1}} \times \mathbb{Z}_{p_2^{\alpha_2}} \times \dots \times \mathbb{Z}_{p_n^{\alpha_n}}$, p_i can be repeated.

Def 5.2 $A = \{g_i | i \in I\}$ The smallest subgroup G containing A is called subgroup generated by A , denote $\langle A \rangle$. if $|A| < \infty$, then is finitely generated.

Thm 5.2 (5.1, Finite generated) $G \cong \mathbb{Z}_{p_1^{\alpha_1}} \times \mathbb{Z}_{p_2^{\alpha_2}} \times \dots \times \mathbb{Z}_{p_n^{\alpha_n}} \times \mathbb{Z} \times \dots \times \mathbb{Z}$

Def 5.3 Actions, let X be a set and G be a group. An action of G on X is a map $G \times X \rightarrow X$, $(g, x) \rightarrow gx$ satisfying (i) $ex = x \forall x \in X$, (ii) $(g_1 g_2)x = g_1(g_2 x) \forall x \in X, g_1, g_2 \in G$

Def 5.4 we say x is G -equivalent to y if $\exists g$ s.t. $gx = y$

(denote $x \sim_G y$) it is equivalent relation.

Def 5.5 if X is a G -set, then each partition of X associated with G -eq is called orbit of X under G . denote O_x with orbit containing x .

Def 5.6 X is G -set and $g \in G$. Fixed point set of $g := X_g = \{x \in X | gx = x\} \forall x \in X$, Stabilizer subgroup of $x := G_x = \{g \in G | gx = x\}$.

Thm 5.3 $G.X$ finite, then $|O_x| = [G : G_x]$

Def 5.7 let X be finite G -set. $X_G := \{x \in X | gx = x, \forall g \in G\}$ as the set of fixed pts. $|X| = |X_G| + \sum_i |O_{x_i}|$ (O_{x_i} distinct) if G act itself by conjugation G_G is the center $Z(G) = \{x | gx = xg \forall g \in G\}$ $|G| = |Z(G)| + [G : G_{x_1}] + \dots + [G : G_{x_k}]$

Thm 5.4 if $|G| = p^r$ (p prime) then G has a nontrivial center.

Cor 5.1 if $|G| = p^2$, then it is Abelian

Thm (hw3) $H \trianglelefteq G$. if G is cyclic/abelian, so is G/H

Thm 6.1 (Cauchy) if a prime p divides $|G|$ ($p || |G|$), then the group G must contain a subgroup of order p .

Thm 6.2 (Sylow I) if $p^r || |G|$ with p prime, then the group G must contain a subgroup of order p^r .

Def 6.2 (see Def 5.1 also) a Sylow p -subgroup P of a group G is the maximal p -subgroup of G .

Def 6.3 let $\mathcal{S}$ be the collection of all subgroup of G . For any subgroups $H, \mathcal{S}$ is a H -set by conjugation $h \cdot K \mapsto hKh^{-1}$. $N(H) = \{g \in G | g \cdot H = H\}$ is a subgroup of G called normalizer H in G . $H \trianglelefteq N(H)$

Thm 6.3 (Sylow II) let G be a finite group and p a prime s.t. $p || |G|$. then all Sylow p -subgroups of G are conjugate. i.e. if P_1, P_2 are Sylow p -subgroups, $\exists g \in G$ s.t. $gP_1g^{-1} = P_2$.

Thm 6.4 (Sylow III) let G be a finite group and p a prime dividing $|G|$. Then the number of Sylow p -subgroup of G satisfy $n_p \equiv 1 \pmod{p}$ and $n_p || |G|$; furthermore, $n_p || |G|/p^r$ (denoted n_p).

Cor 6.2 if p, q are primes and $p < q$, then every group of order pq has only one normal subgroup of order q . if $q \not\equiv 1 \pmod{p}$, then G is cyclic.

Remark 6.1 if $n_p = 1$, then the single Sylow p -group is a normal subgroup of G . unless $|G| = p$, the group G is not simple. if G is simple, then $|G| \nmid n_p!$. Note: intersection of Sylow p -subgroup is $\{e\}$.

Def 6.4 A non-empty set R is a ring if $+, \cdot$ are closed and $\forall a, b, c \in R$: (i) $(R, +)$ is an Abelian group, and denote $e = 0, r^{-1} = -r$. (ii) $(ab)c = a(bc)$ (iii) $a(b+c) = ab+ac$, $(a+b)c = ac+bc$.

Def 6.5 (types of Ring) (i) if $\exists 1_R \in R$ s.t. $1_R \cdot a = a \cdot 1_R = a \forall a \in R$, then R is a ring with identity. (ii) if $ab = ba \forall a, b \in R$, then R is a commutative ring (iii) is an integral domain if $\forall a, b \in R, ab = 0$ implies $a = 0$ or $b = 0$. (iv) is a division ring if every nonzero element in R is a unit (i.e. has unique inverse a^{-1} s.t. $aa^{-1} = a^{-1}a = 1_R$). (v) a Field is a commutative division ring. (v) $\Rightarrow$ (iii), (iii) $\Leftrightarrow$ if $a \neq 0, ab = ac$ iff $b = c$ (cancelation law)

Def 6.6 $a \neq 0 \in R$ is a zero divisor if there is a $b \neq 0 \in R$ s.t. $ab = 0$.

Prop 6.3 if R is a ring and $a, b \in R$, then (i) $a0 = 0a = 0$, $a(-b) = (-a)b = -ab$.

Def 6.7 $S \subset R$ is a subring if (i) $S \neq \emptyset$, (ii) $\forall r, s \in S, r-s \in S$, (iii) $\forall r, s \in S, rs \in S$.

Thm 6.5 every finite integral domain is a field.

Def 6.8 the Characteristic of a ring R is the smallest positive integer n s.t. $nr = 0 \forall r \in R$. denote by $\text{char } R = n$, if does not exist, $\text{char } R = 0$. and 1_R is identity

Prop 6.4 let R is a ring with identity, if 1_R has order n , then $\text{char } R = n$ (ex. $\text{char } \mathbb{Z}_p = p$)

Thm 6.6 characteristic of a integral domain is either prime or zero.

Def 7.1 let R, S be rings, a map ϕ is a ring homomorphism $R \rightarrow S$ if $\phi(a+b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$. In addition, if ϕ is bijective, then is a isomorphism. $\text{Ker}\phi := \{r \in R | \phi(r) = 0\}$

Prop 7.1 let $\phi : R \rightarrow S$ is a ring homomorphism. (i) if R is a commutative ring, so is $\phi(R)$ (ii) $\phi(0) = 0$ (iii) if ϕ is onto, then $\phi(1_R) = 1_S$. (iv) if R is a Field and $\phi(R) \neq \{0\}$, then $\phi(R)$ is also a field.

Def 7.2 An ideal in a ring R is a subring I s.t. $rI \subset I, Ir \subset I, \forall r \in R$. if I is a ideal contain 1_R , then $I = R$.

Def 7.3 if R is a commutative ring with identity, then $\langle a \rangle = \{ar | r \in R\}$ is an ideal and is principal ideal.

Thm 7.1 every ideal of $\mathbb{Z}$ is principal.

Prop 7.2. if ϕ is a ring homomorphism: $R \rightarrow S$, then $\text{Ker}\phi$ is an ideal of R .

Thm 7.2 R/I is a ring with multiplication $(r+I)(s+I) = rs+I$ (factor ring)

Thm 7.3 $\phi : R \rightarrow R/I, \phi(r) = r+I$ is canonical homomorphism, $\text{Ker}\phi = I$.

Thm 7.4, 5, 6 similar isomorphism thm to groups

Def 7.4 an ideal M of ring R is maximal if $M \subseteq I \subseteq R$, then either $I = M$ or $I = R, M \neq R$.

Thm 7.7 let R be commutative ring with identity and M an ideal of R . Then M is maximal iff R/M is a field.

Def 7.5 an ideal $P \subset R$ is a prime ideal if $\forall ab \in P$, either $a \in P$ or $b \in P$.

Thm 7.8 P is a prime ideal in commutative ring R with identity iff R/P is an integral domain ($1_R \neq 0$).

Cor 7.1 maximal ideal in a commutative ring with identity is also prime ideal.

Thm 8.1 if R is a commutative ring with identity, so is $R[x]$.

Def 8.3 A nonconstant polynomial $f(x) \in F[x]$ is irreducible over the field F if $f(x)$ cannot be expressed as a product of two polynomials in $F[x]$ with degrees smaller than that of $f(x)$.

Lemma 8.1 if $p(x) \in \mathbb{Q}[x]$, then $p(x) = \frac{r}{s}(a_0 + \dots + a_n x^n)$ r, s, a_i integers and r, s coprime, a_i coprime.

Thm 8.4 (Gauss) let $p(x) \in \mathbb{Z}[x]$ with leading coefficient 1 (monic), if $p(x) = \alpha(x)\beta(x)$ where $\alpha(x), \beta(x) \in \mathbb{Q}[x]$, then $p(x) = a(x)b(x)$, $a(x), b(x) \in \mathbb{Z}[x]$, $\deg a(x) = \deg \alpha(x)$, $\deg b(x) = \deg \beta(x)$.

Cor 8.4 $p(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$. if $p(x)$ has zeros in $\mathbb{Q}$, then also has a zero $\alpha \in \mathbb{Z}$, and $\alpha | a_0$.

Thm 8.5 (Eisenstein) $f(x) = a_n x^n + \dots + a_0 \in \mathbb{Z}[x]$. if $p | a_i$ for $i = 0, 1, \dots, n-1$, $p \nmid a_n$ and $p^2 \nmid a_0$ with p prime, then $f(x)$ is irreducible over $\mathbb{Q}[x]$.